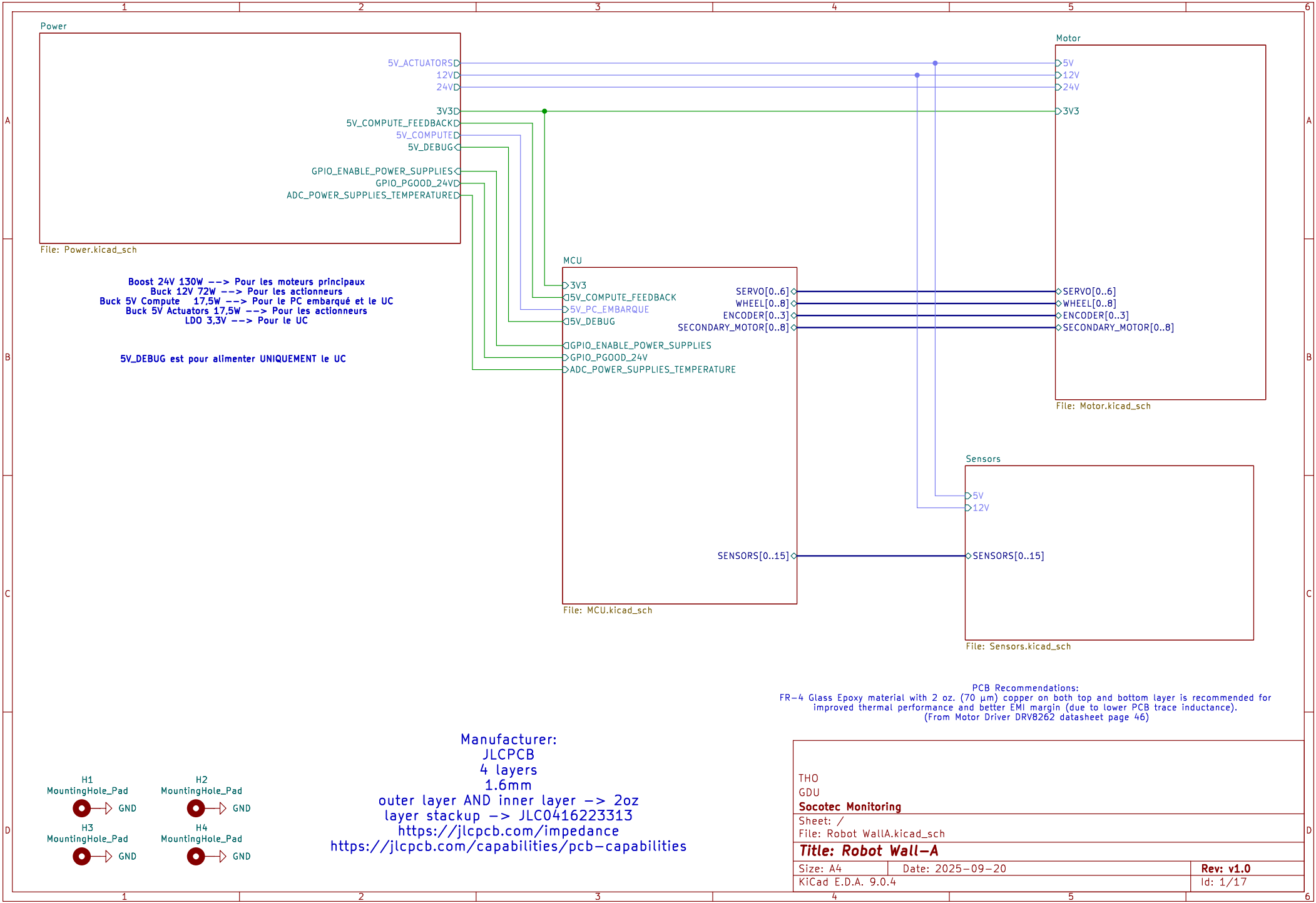
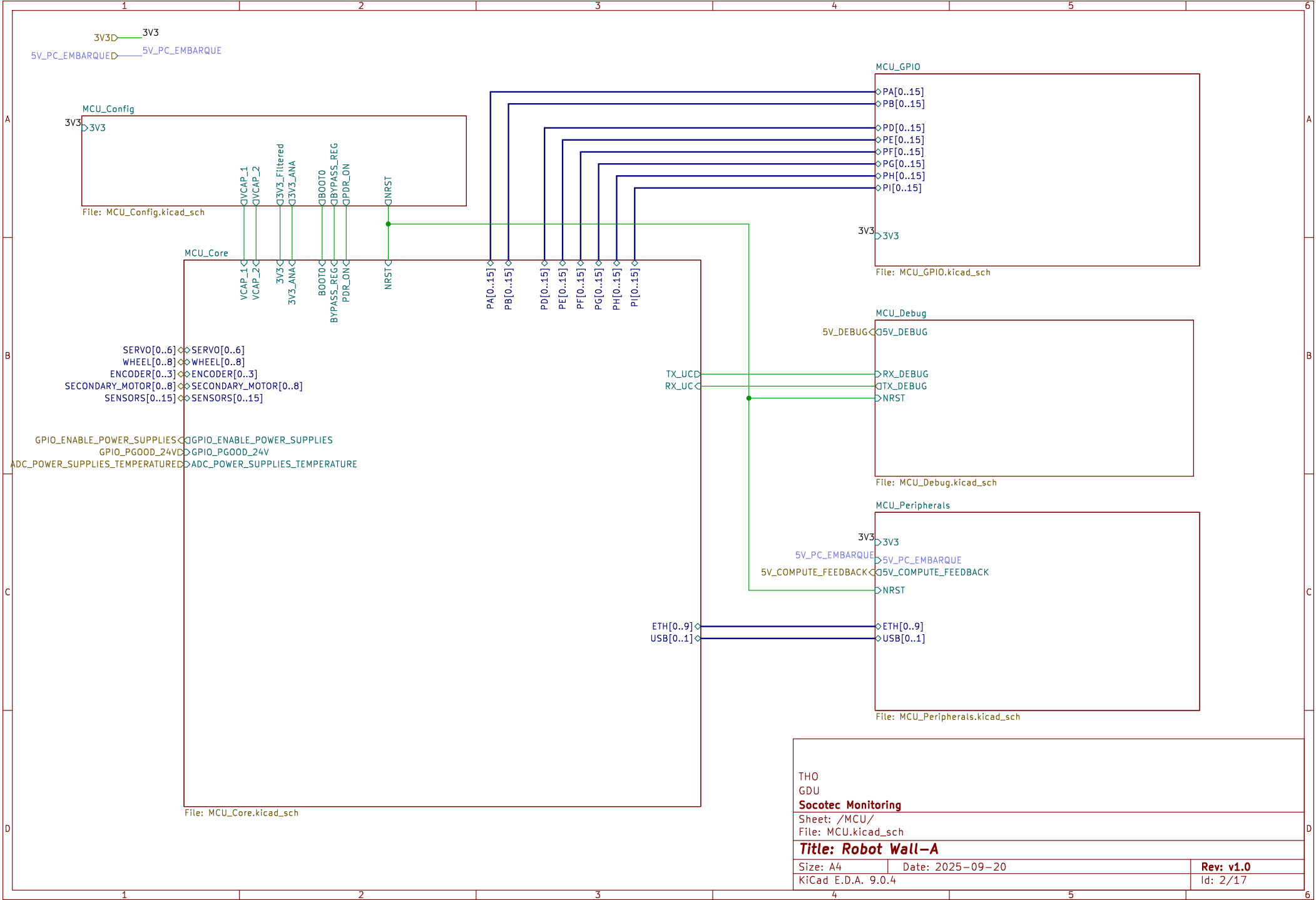
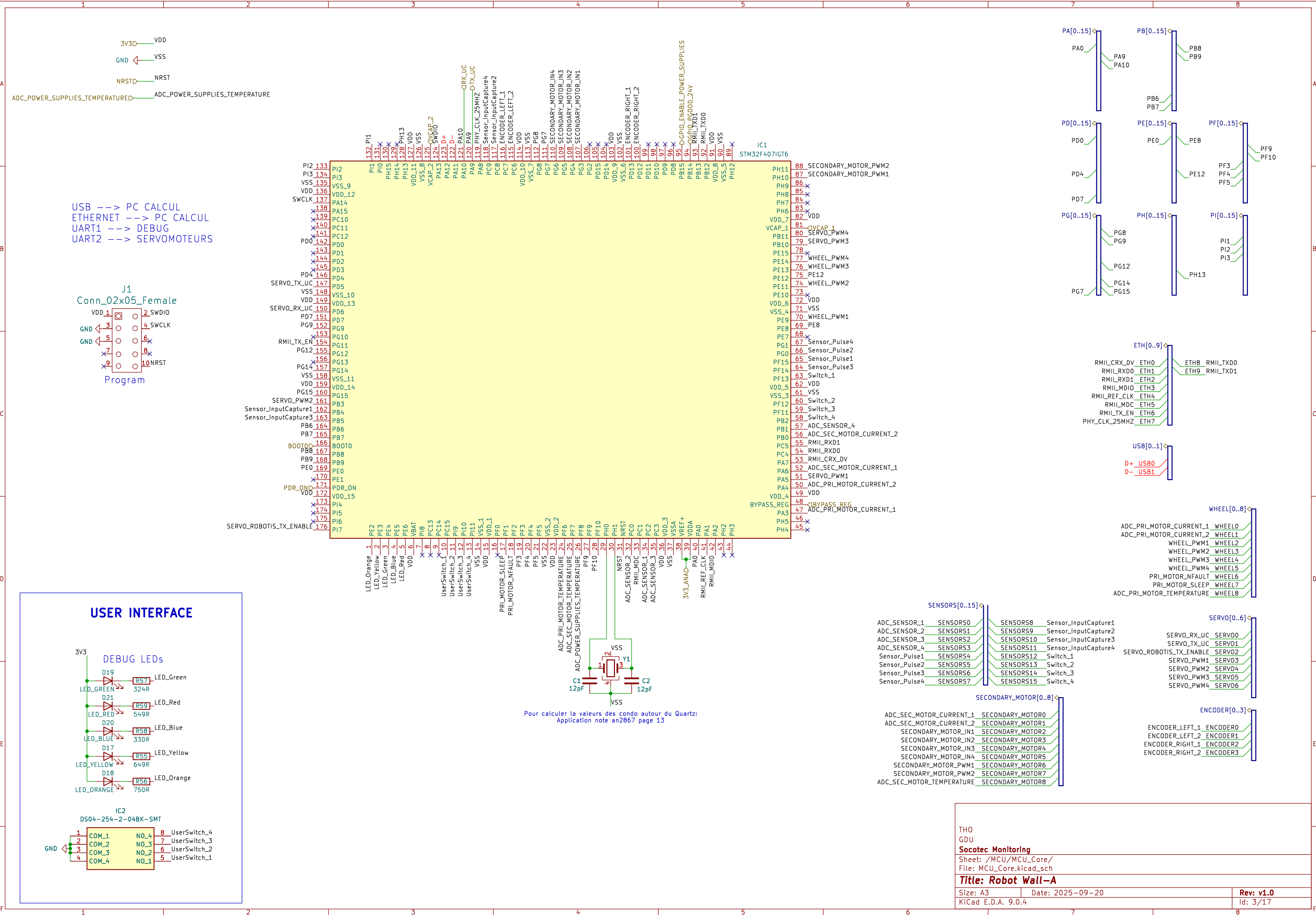


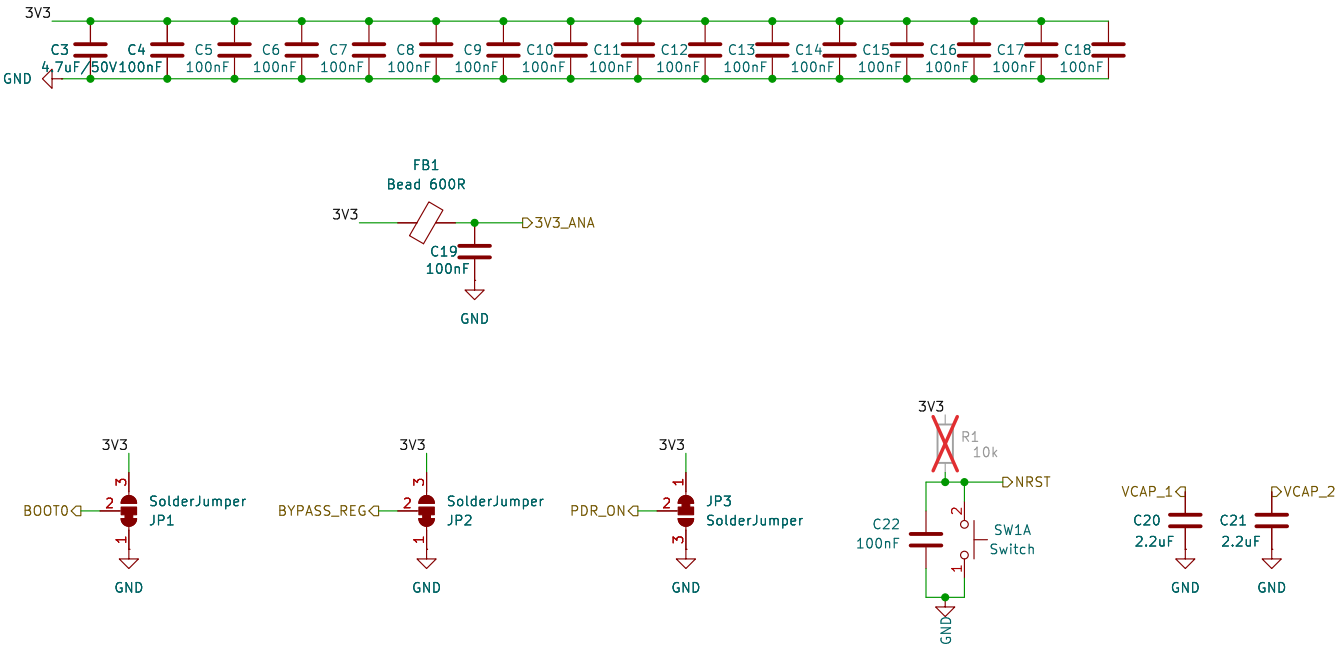






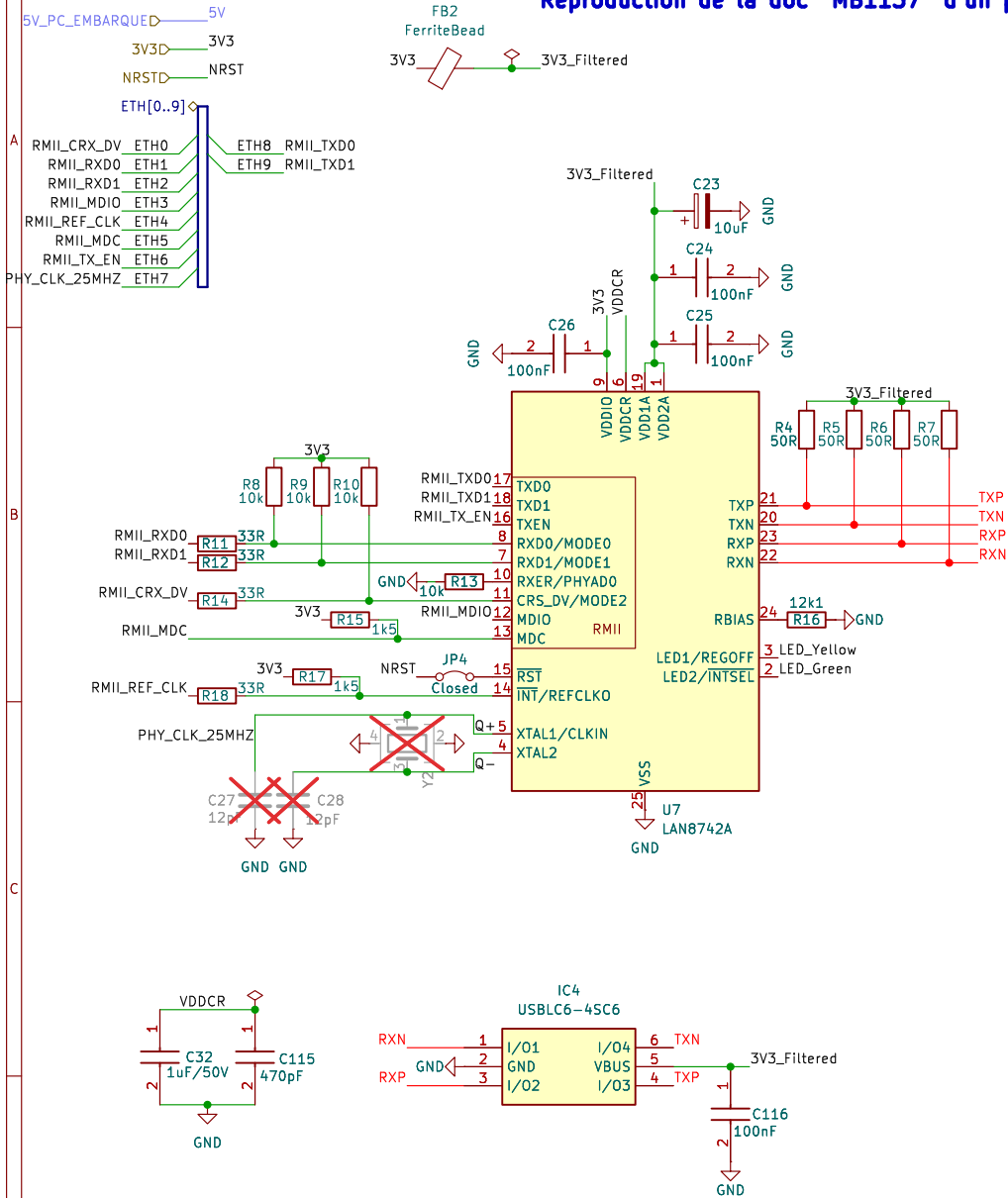
Must be as close as possible from μC

3V3 $\rightarrow$ 3V3_Filtered

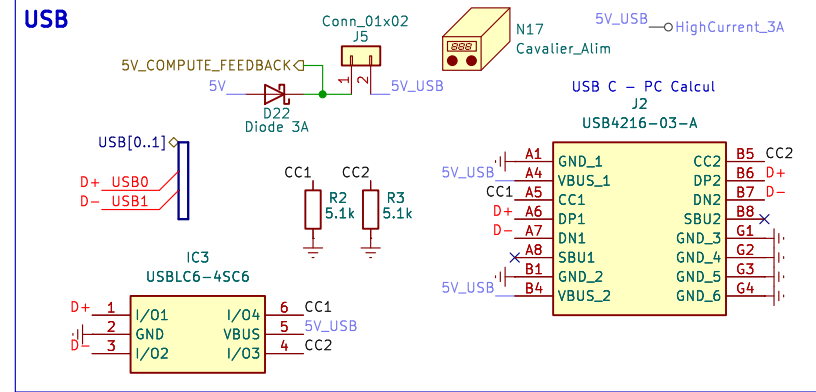


TH0		
GDU		
Socotec Monitoring		
Sheet: /MCU/MCU_Config/		
File: MCU_Config.kicad_sch		
Title: Robot Wall-A		
Size: A4	Date: 2025-09-20	Rev: v1.0
KiCad E.D.A. 9.0.4	Id: 4/17	

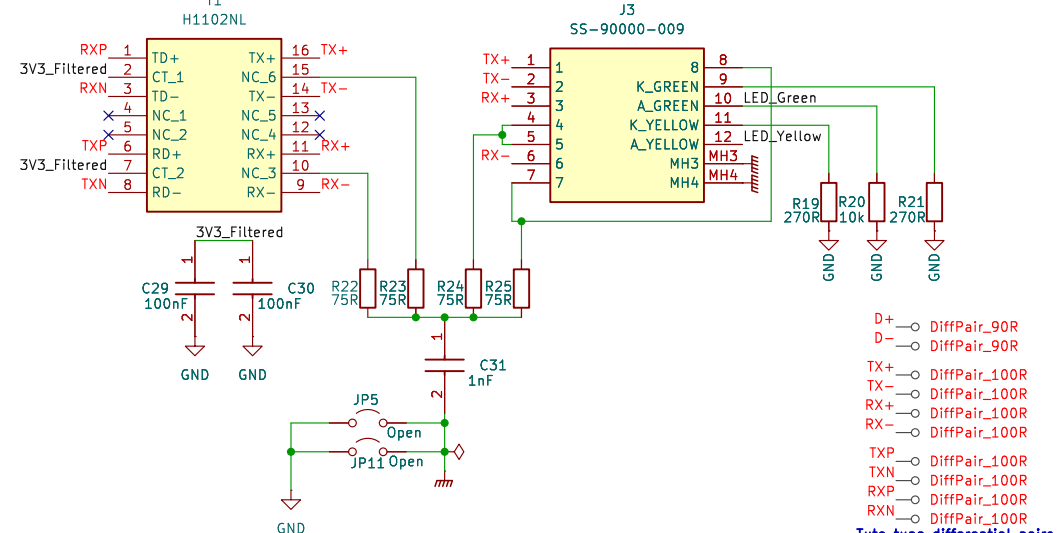
Reproduction de la doc "MB1137" d'un µC STM32 (pdf dans le git, dossier Docs)



USB



TX peut etre connecté sur RX car le module est compatible auto-MDIX



Tuto tune differential pairs
<https://www.youtube.com/watch?v=M13QxtPvXYt&t=798s>

TH0
GDU

Socotec Monitoring

Sheet: /MCU/MCU_Peripherals/
File: MCU_Peripherals.kicad_sch

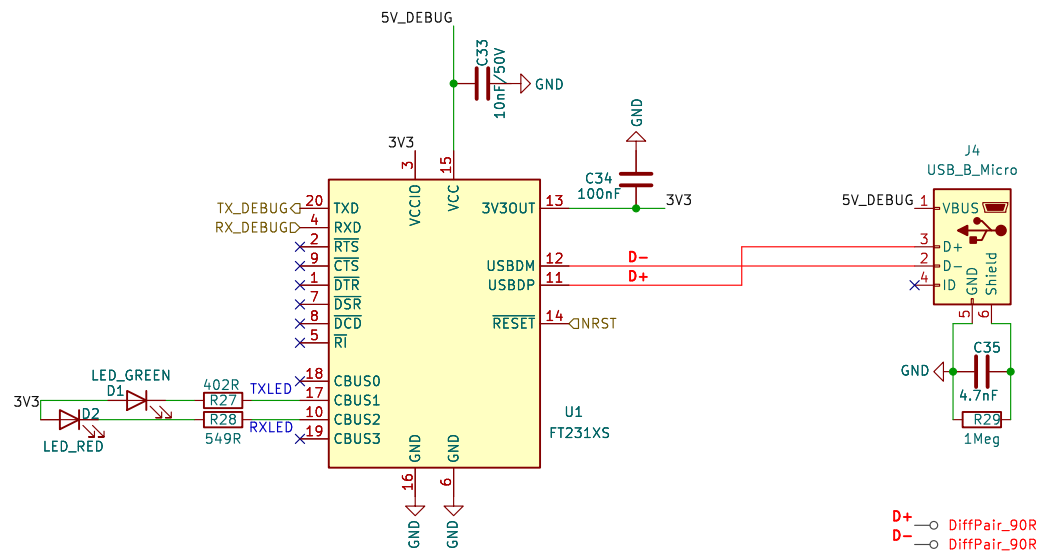
Title: Robot Wall-A

Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.4

Rev: v1.0

Id: 5/17



TH0
GDU

Socotec Monitoring

Sheet: /MCU/MCU_Debug/
File: MCU_Debug.kicad_sch

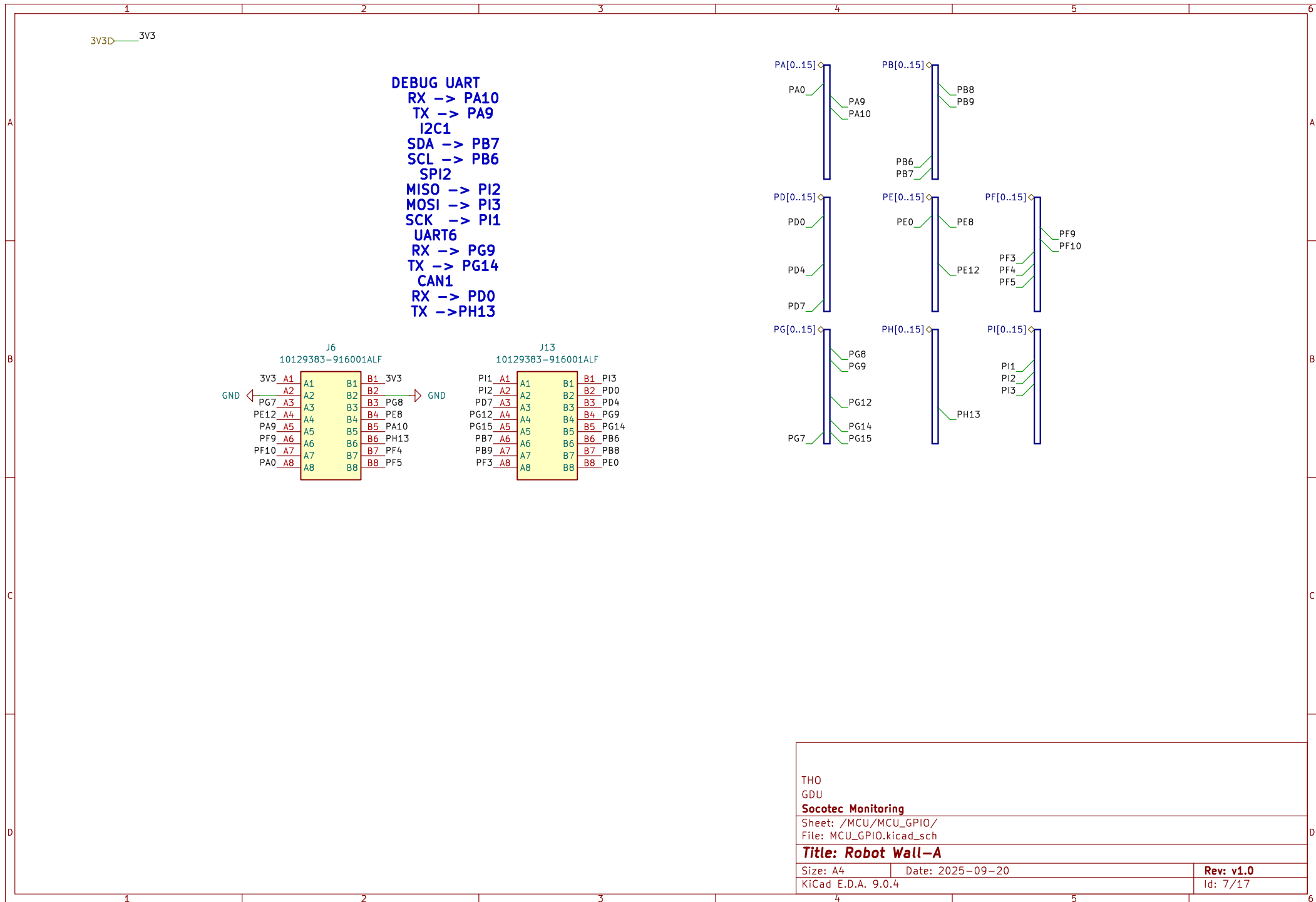
Title: Robot Wall-A

Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.4

Rev: v1.0

Id: 6/17



TH0
GDU

Socotec Monitoring

Sheet: /MCU/MCU_GPIO/
File: MCU_GPIO.kicad_sch

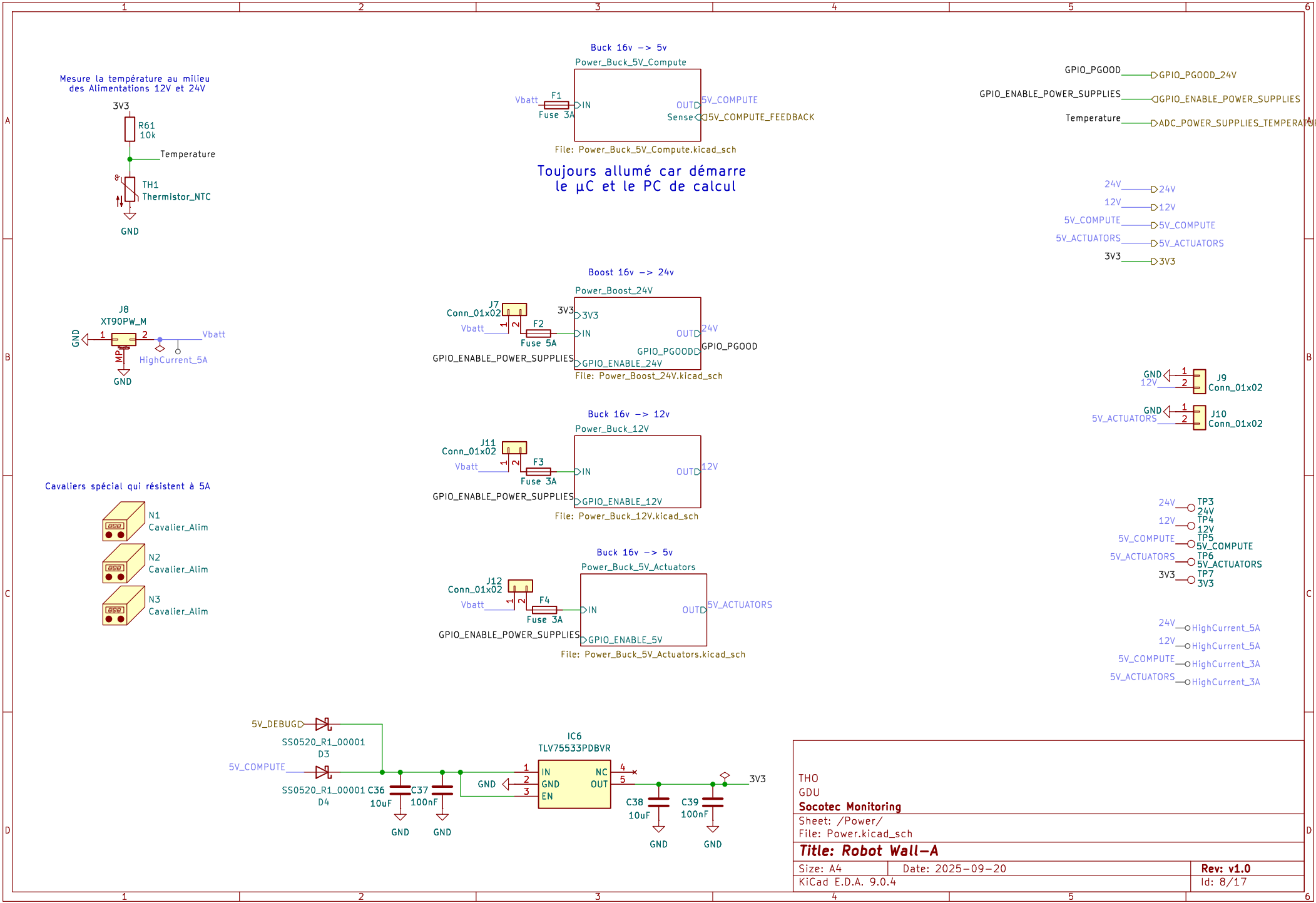
Title: Robot Wall-A

Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.4

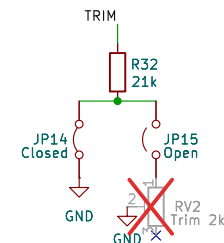
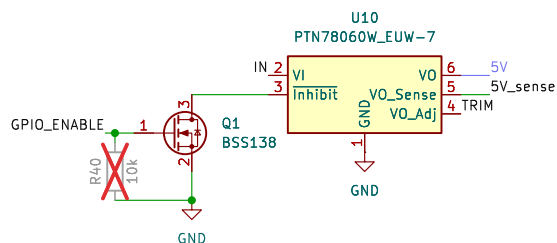
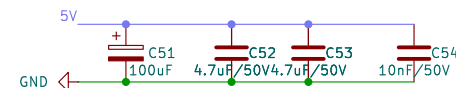
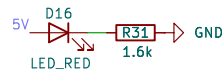
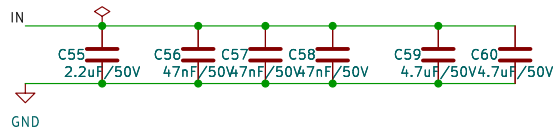
Rev: v1.0

Id: 7/17



IND IN

5V OUT
5V_sense Sense



TH0
GDU

Socotec Monitoring

Sheet: /Power/Power_Buck_5V_Compute/
File: Power_Buck_5V_Compute.kicad_sch

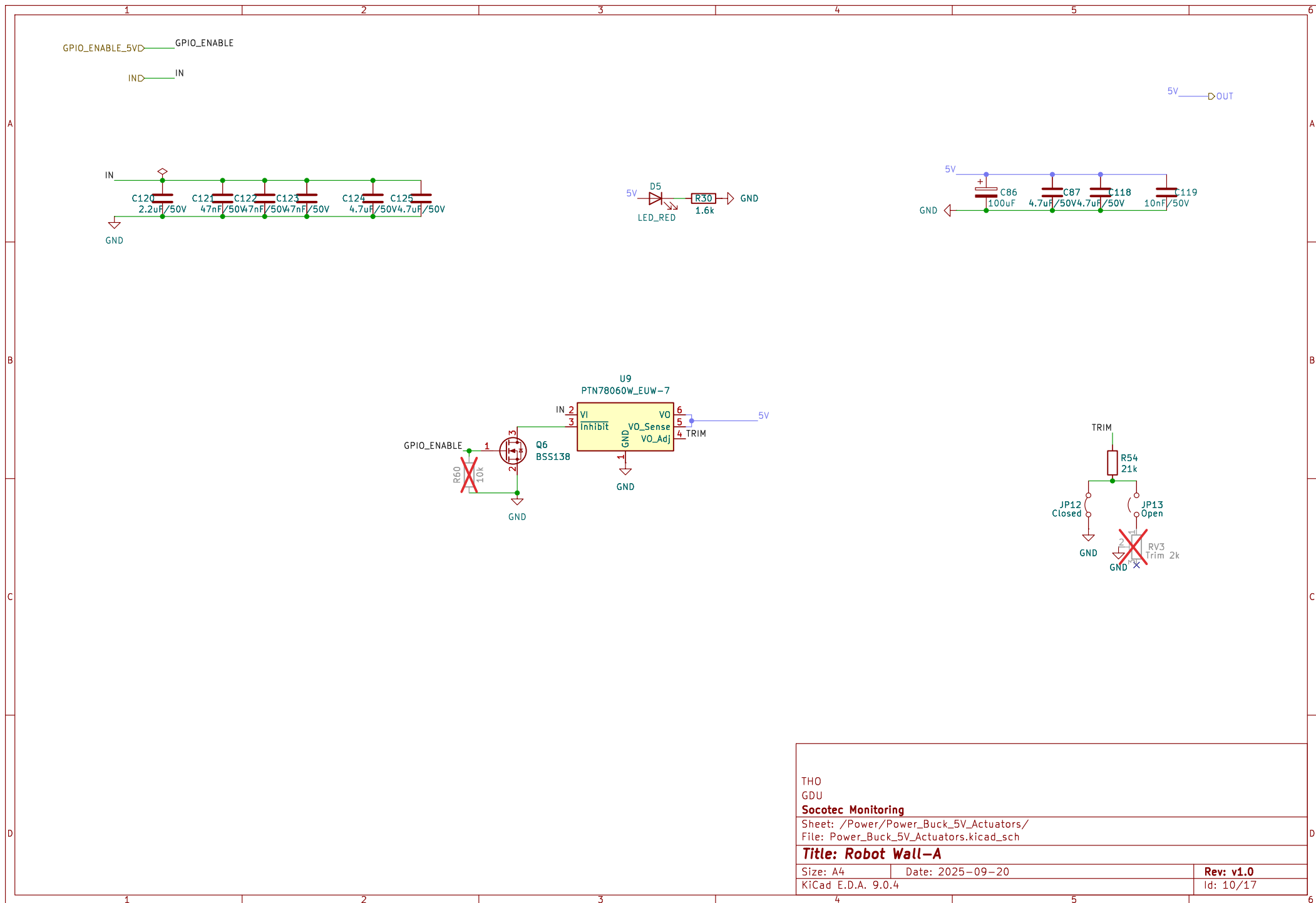
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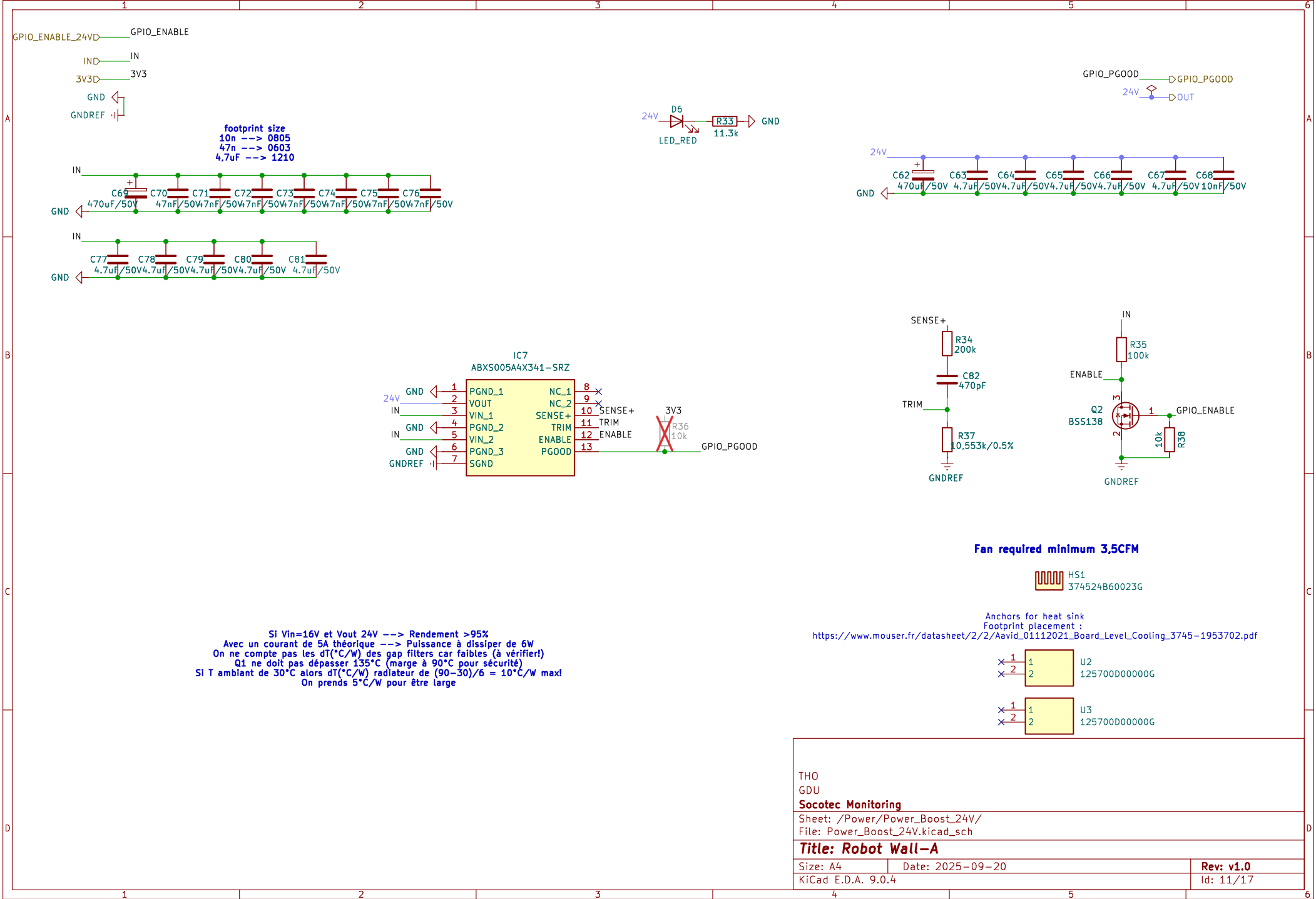
Size: A4 Date: 2025-09-20

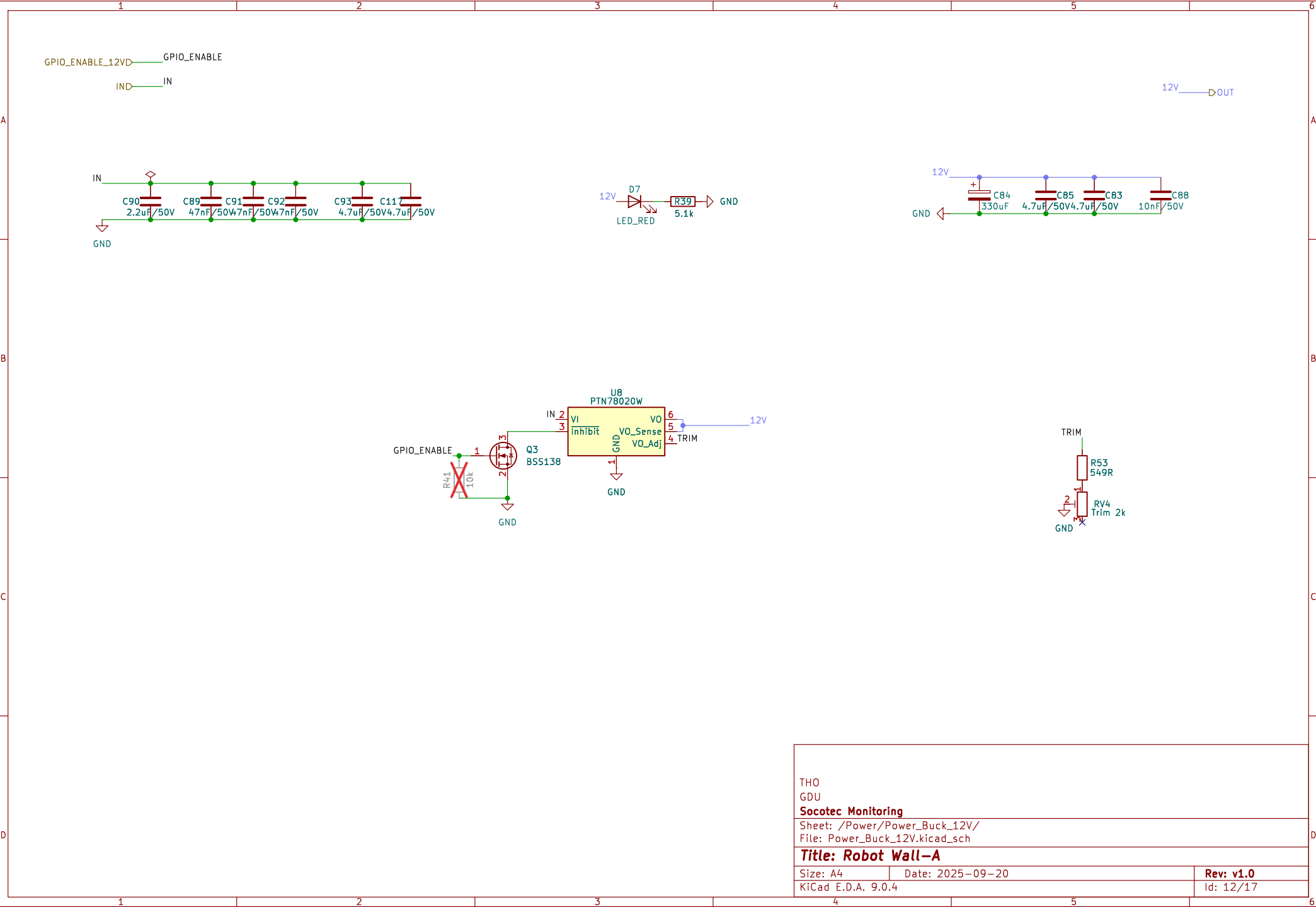
KiCad E.D.A. 9.0.4

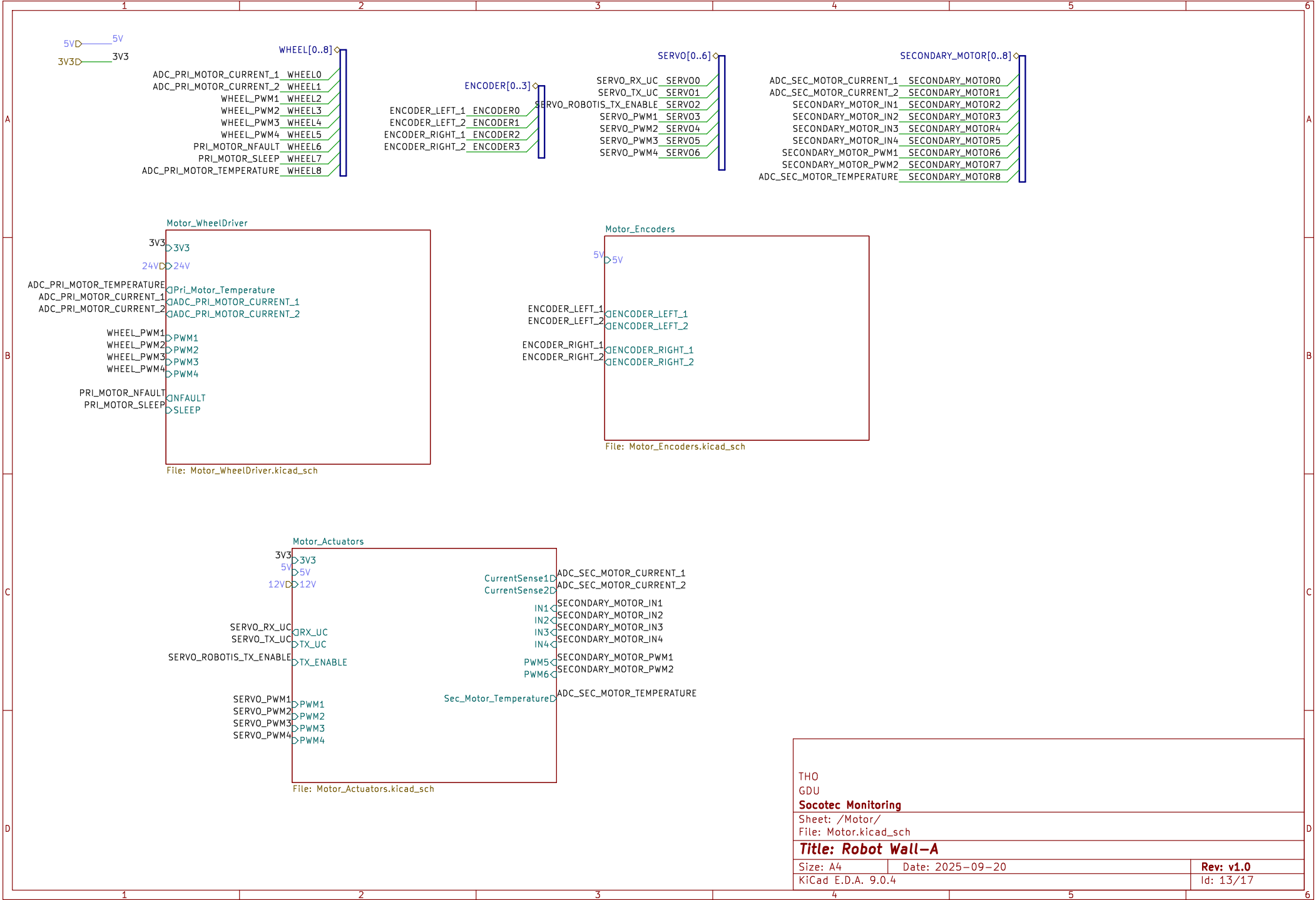
Rev: v1.0

Id: 9/17







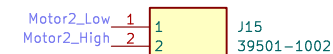
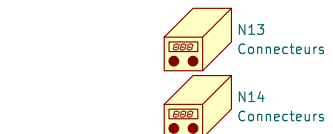
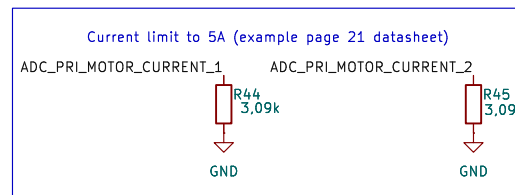
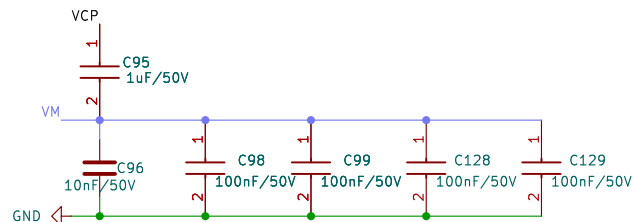
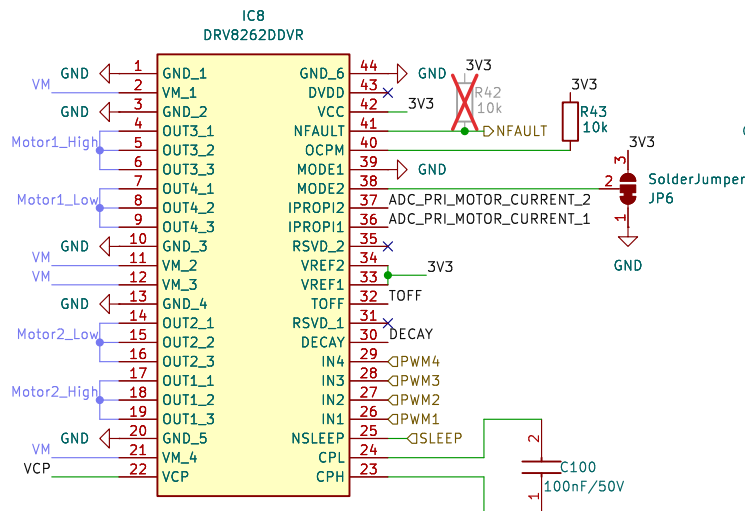


3V3D — 3V3
24VD — VM



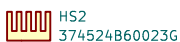
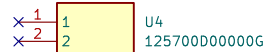
Default Configuration:
OCPM = 1 --> Over Current Protection error = Retry
MODE = 0 --> PH/EN mode
Toff = 1 --> Waiting time between change = 16µs
Decay = HZ --> Auto detect mode

ADC_PRI_MOTOR_CURRENT_1 —> ADC_PRI_MOTOR_CURRENT_1
ADC_PRI_MOTOR_CURRENT_2 —> ADC_PRI_MOTOR_CURRENT_2



ATTENTION!!
Pour simplifier le routage,
les phases du moteurs on été inversées.
Attention au câblage!

https://www.mouser.fr/datasheet/2/2/Aavid_01112021_Board_LeveL_Cooling_3745-1953702.pdf

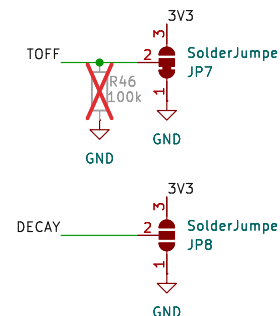
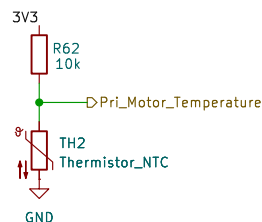


Fan required minimum 3,5CFM

2x Transmotec PD4266-24-17-BFEC
24V / 2,1A nominal
4,2A nominal total

Power loss calculation and dissipator estimation
(page 33 to 35 datasheet)
Calculations already done in datasheet (same example with 4A)
Conclusion: Dissipator with less than 4°C/W

Heat sink possibilities:
WAVE-35-12 with anchors 125700D00000G --> Fan required minimum 3CFM
374524B60023G with anchors 125700D00000G --> Fan required minimum 3,5CFM
567-45AB with screws --> Fan required minimum 4CFM



Motor1_High —> HighCurrent_5A
Motor1_Low —> HighCurrent_5A
Motor2_High —> HighCurrent_5A
Motor2_Low —> HighCurrent_5A

TH0
GDU

Socotec Monitoring

Sheet: /Motor/Motor_WheelDriver/
File: Motor_WheelDriver.kicad_sch

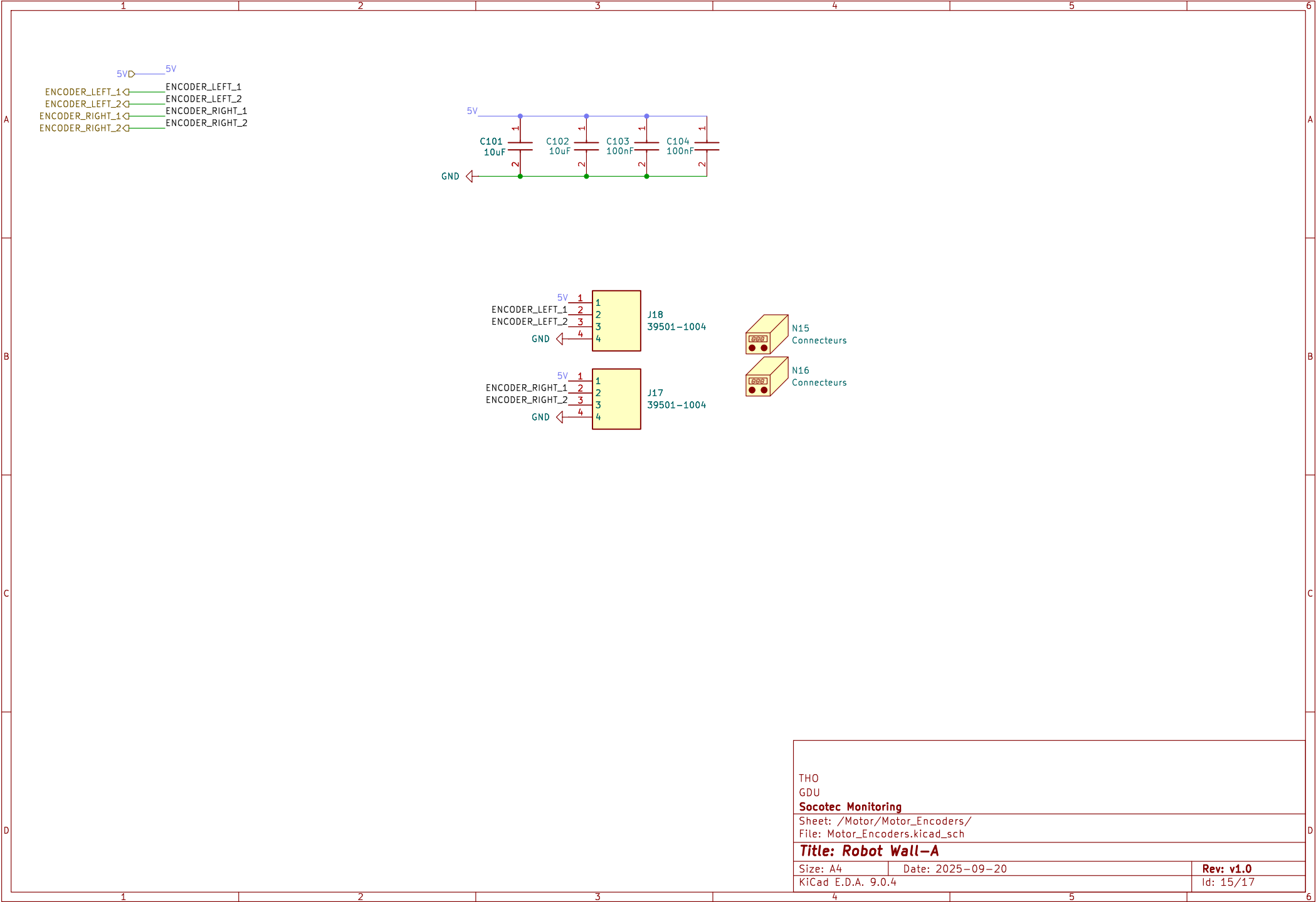
Title: Robot Wall-A

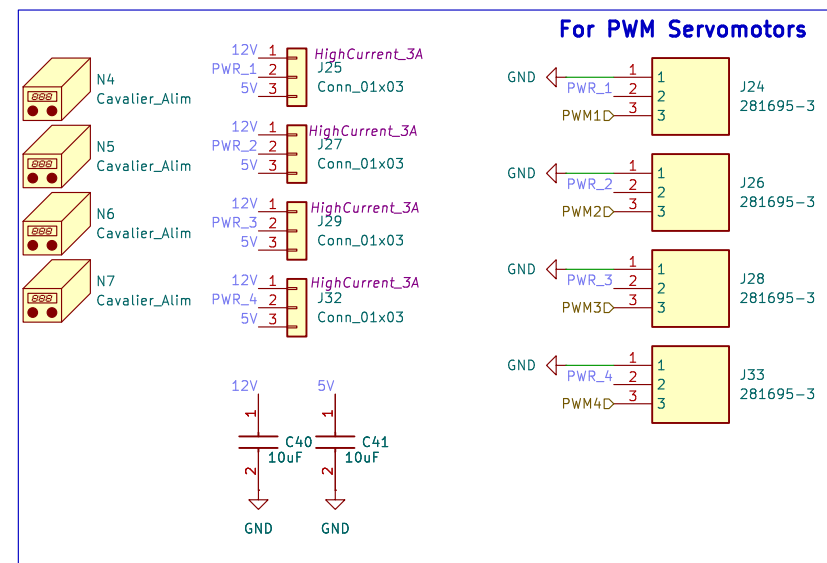
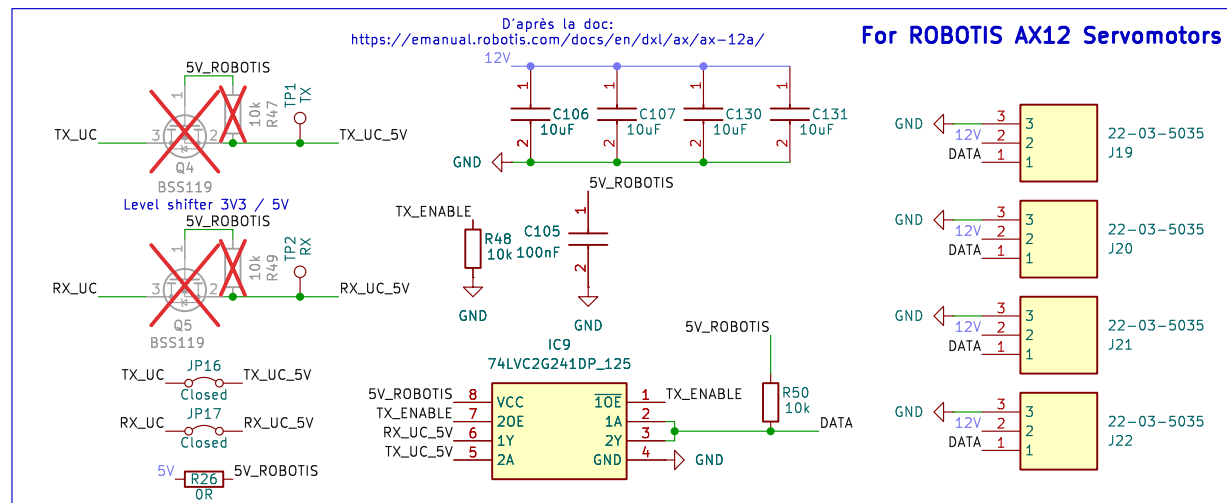
Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.4

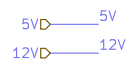
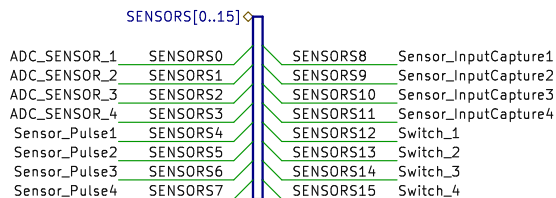
Rev: v1.0

Id: 14/17



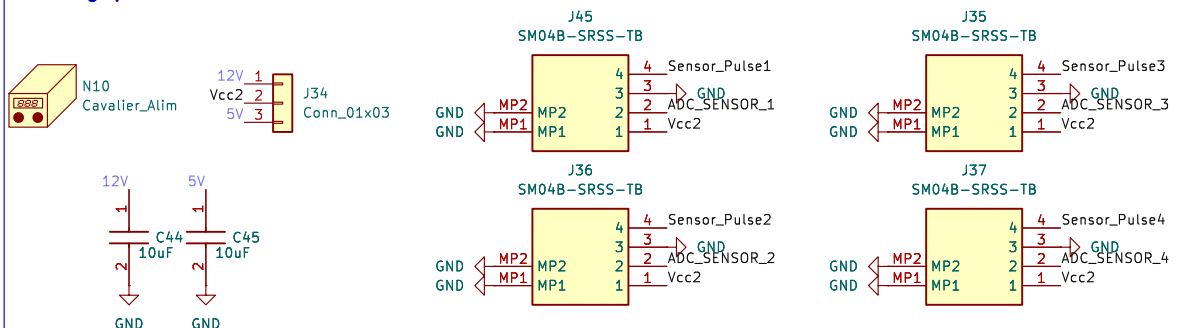


Id: 16/17



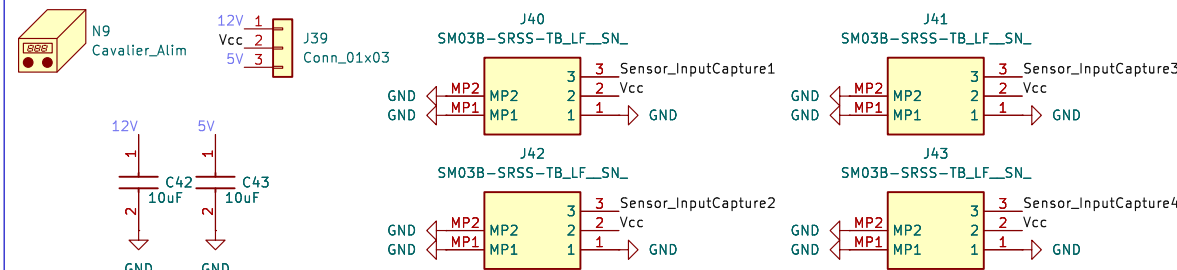
Capteurs de distance

Pour capteur B5W-LB2101-1 analogique 5V

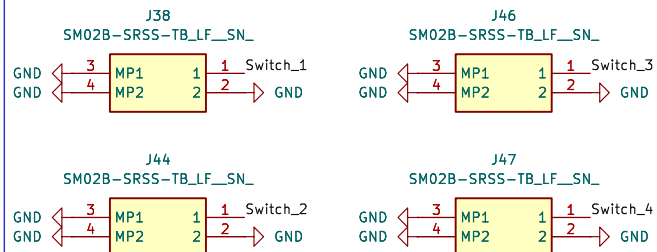


https://omronfs.omron.com/en_US/ecb/products/pdf/en-b5w_lb_series_users_manual.pdf

Pour capteur Pololu <https://www.pololu.com/product/5464>



Interrupteurs standard



TH0
GDU

Socotec Monitoring

Sheet: /Sensors/
File: Sensors.kicad_sch

Title: Robot Wall-A

Size: A4 Date: 2025-09-20

KiCad E.D.A. 9.0.4

Rev: v1.0

Id: 17/17